



Verification Report:

Name of Verification company:	Date of the issue:
Scientific Certification Systems (SCS)	April 28, 2011
Report Title:	Approved by:
Verification Report: Darkwoods Forest Carbon Project	Robert J. Hrubes
Client:	Project Title:
3GreenTree Ecosystem Services Ltd. ERA Ecosystem Restoration Associates Inc. Nature Conservancy of Canada	Darkwoods Forest Carbon Project
Summary:	
The Darkwoods Forest Carbon Project is an “Improved Forest Management” project implemented by Nature Conservancy of Canada that results in greenhouse gas (GHG) emission reductions and removals through the avoidance of emissions due to logging in the baseline scenario. The project utilized the approved Verified Carbon Standard (VCS) methodology VM0012-Improved Forest Management on Privately Owned Properties in Temperate and Boreal Forests (LTPF) Version 1.0. SCS was hired as the verifier of record for this project, while Rainforest Alliance served as the validator. The verification of the project included an assessment of monitoring activities and data collected, evaluation of compliance to the VCS standards and to the methodology selected, and an assessment of deviations to the monitoring plan and methodology. The verification process also involved an in-depth assessment of the ex-post calculation of VCUs generated by the project for the reporting period of April 1, 2008 (project initiation) through December 31st 2010. Finally, the audit also included the first verification of the “Risk Assessment and Market Leakage Assessment” as required by the VCS through the Double Approval Process. The audit conducted by Scientific Certification Systems included desk reviews of the project and supporting documentation, a site visit, and interviews with project proponents and technical service providers. It also included field work in which a sample of forest inventory plots were checked by the audit team. During the audit, an iterative exchange of requests for new information and corrections of non-conformances took place between the verifier and the project proponent. At this time, all nonconformities have been adequately resolved. The auditors conclude that the project meets all relevant requirements of the Verified Carbon Standard. SCS verified that the project generated 849,016 t CO₂ equivalent net emissions reductions during the present reporting period.	
Work carried out by:	Number of pages:
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1 Introduction

1.1 Objective

- Evaluate the verification scope, including the GHG project and baseline scenario; GHG sources, sinks, and reservoirs; and the physical infrastructure, activities, technologies and processes of the GHG project.
- Evaluate the monitoring plan and develop conclusions regarding the monitoring methodology and the collection and archiving of data relevant to GHG emissions estimation and baseline emissions.
- Assess conformance of the proposed Agriculture, Forestry and Other Land Use (AFOLU) Project Elements with the VCS Standard and VCS Program Guidelines;
- Evaluate the proposed AFOLU Project Elements based on guidance given under the VCS Program and VCS Program Guidelines to provide a risk assessment and market leakage assessment, which will impact the issuance of VCUs
- Provide the first independent assessment of the double approval process for the proposed AFOLU Project Elements.
- Evaluate the calculation of GHG emissions, including appropriateness of source, sink, and reservoirs; the correctness and transparency of formulae and factors used; assumptions related to estimating GHG emission reductions; and uncertainties.
- Determine if there have been any significant changes to the project procedures or criteria since the project validation.
- Determine if there have been any significant changes in the project and baseline emissions, removals, emission reductions and removal enhancements since the project validation.
- Develop conclusions based on verification standards, submitting any corrective action requests, as applicable.

1.2 Scope and Criteria

SCS assessed the completeness of the Project Monitoring Report to ensure that all requirements of the VCS standards and applicable methodology elements have been addressed. SCS assessed whether or not the Monitoring Report respects the principles of the VCS standards. The Project Design Document (PDD) was referenced during this assessment, but was not itself assessed, as that task was performed during the validation audit, carried out by another validation and verification body. The appropriateness of calculations and parameters presented in the validated PDD were assumed to have been already assessed at validation, and were not additionally assessed in this review.

Assessment included but was not limited to evaluation of the project implementation, monitoring, and the calculation of GHG emissions.

SCS assessed whether the project itself meets all of the requirements laid out in the VCS standards and is consistent with the PDD.

Criteria from the following standards and documents were used to assess the project:

- VCS 2007.1
- VCS Program Guidelines 2007.1
- VCS Guidance for Agriculture, Forestry and Other Land Use Projects (AFOLU)
- VCS Program Normative Document: Double Approval Process, Version 1.1
- Tool for AFOLU Non-Permanence Risk Analysis and Buffer Determination
- VCS Tool for AFOLU Methodological Issues
- VCS Program Update: Updates to the Tool for AFOLU Methodological Issues and Guidance for AFOLU Project: Insignificant Emissions Sources and Pools, Carbon Pools, Avoided Planned Deforestation, Definition of Mosaic and Frontier Deforestation, Market Leakage
- Methodology: VM0012-Improved Forest Management on Privately Owned Properties in Temperate and Boreal Forests (LtPF) Version 1.0.

The assessment was performed using the client-supplied Monitoring Report, PDD, and other supporting documentation.

1.3 VCS project Description

Page 4 of the PDD states: *“The Nature Conservancy of Canada (NCC) acquired the fee simple 54,792 ha (135,394 acre) Darkwoods Property near Creston, BC from the Pluto Darkwoods Corporation in April of 2008, with the objective of managing the land for ecological conservation objectives. NCC is developing a 100 year VCS IFM-LtPF forest carbon project with a start date of the acquisition date to provide carbon finance as a significant part of acquisition financing and funding for ongoing property management and ownership costs.*

The Darkwoods Forest Carbon Project achieves net GHG emission reductions and removals through the avoidance of emissions due to logging in the baseline scenario. The Darkwoods property was being sold by the previous owner, Pluto Darkwoods on a bid basis based on a sales price from a formal property and timber valuation/appraisal similar to the baseline scenario. The most plausible baseline scenario is a market-driven acquirer who implements a 15 year depletion of current mature timber stocks to provide a reasonable internal rate of return on investment. Under the baseline scenario, a 100 year uneven harvest schedule is implemented with the typical regional practice of clearcut logging with minimum legal requirements for private forestlands in B.C. and comparable regional practices.

The project scenario is conservation management activities, wherein NCC undertakes the carbon project and ecosystem protection and enhancement activities. The project scenario anticipates a low level of timber removal as part of conservation management activities for ecosystem/habitat enhancement and risk management.”

1.4 Level of assurance

SCS provides reasonable assurance per section 7.3.1 of the VCS standard (“VCS 2007.1”) that the emission reduction estimations for the Darkwoods Forest Carbon Project, located near Creston, British Columbia are conservative and meet the VCS criteria and approved methodology, VM0012-Improved Forest Management on Privately Owned Properties in Temperate and Boreal Forests (LtPF) Version1.0.

To ensure complete transparency, SCS has included any clarification or corrective actions that were raised at the end of this validation report.

2 Methodology

2.1 Review of Documents

SCS received and reviewed the PDD and supporting documentation to assess initial conformance with the requirements of the VCS standard. Key factors that impact the reported emission reductions and removals were identified, and a Verification Plan was created to focus on the critical elements presenting potential risk for errors. These elements included inventory data collection and handling, evidence of conformance to the methodology, evidence of conformance to the monitoring plan, calculations, and modeling.

The desk review resulted in findings associated with the time elapsed since the last on-the-ground forest inventory which had not been conducted since 1999. It was the understanding of the verification team that the methodology required monitoring of forest inventory through field measurements at intervals of no greater than every five years. As such it was determined that the project was out of conformance with the methodology. This issue was further discussed during and after the site visit. This report details the findings that led to two Non-Conformity Reports (NCR) and two New Information Requests (NIR), that the project proponents had to respond to in order to allow SCS to provide reasonable assurance that the emission reduction estimations meet the requirements of the VCS and the selected methodology. The client’s satisfactory responses are detailed, along with the verifier’s responses that lead to the closure of the identified NCR/NIRs.

Finally the document was technically reviewed for completeness by an SCS auditor for quality assurance purposes.

2.2 Site Visit

Following the desk review, the auditors conducted an on-site audit of the project area on March 6-12, 2011. The first two days of the site visit involved meeting in the offices of the project proponent to conduct detailed discussions surrounding the PDD and monitoring report, the remote sensing and GIS analysis employed, the calculation of VCUs, and the use of growth and yield models. The site visit was also used to assess the application of the PDD, visit random locations and plots within the project area for reconnaissance and ground-truthing, and to conduct a field review

of the project proponent's sampling methodology. During the site visit, two inventory plots were re-measured by the audit team. The collected data was used to confirm whether proper field measurement techniques were used in project monitoring. Stand typing and general descriptions of the forested area were also confirmed as part of the general inspection of the project area.

2.3 Quantitative Analysis

The third step of the verification process focused on the quantitative analyses undertaken by the Project Proponent to convert the raw inventory data into estimates of standing volume of woody biomass and then to translate volumetric measures into estimated tons of carbon, as presented in the monitoring report. This entailed generating carbon stock estimates from the field data collected by the audit team and comparing it to estimates made by the project proponent. Additionally, calculations made by the project proponent were reviewed by the audit team.

2.4 Final Review and Report Drafting

The last step in the verification process included a final review of the submitted data, and drafting of the VCS Verification Deed of Representation and supporting Verification Report. A draft Verification Deed of Representation and Report was completed based on the results of the verification assessment. The draft report was presented to an internal SCS Technical Reviewer who determined the Verification Opinion to be justified given the evidence presented. The report and opinions were then presented to the project proponent for review and comment.

2.5 Follow-up Interviews

The following personnel were interviewed during the audit process:

Mike Vitt, Managing Director 3GreenTree
Clive Welham, Managing Director 3GreenTree
Brad Seely, 3GreenTree
Tom Swann, Associate Regional VP Nature Conservancy of Canada
Christian Schadendorf, General Manager Darkwoods Forestry
Kathleen McGuinness, Touchstone GIS Consulting
Milt Hamilton, Nanuq Consulting Ltd.
Trevor Anderson, Timberland Consultants

2.6 Resolution of any material discrepancy

Throughout the validation/verification process, there was an iterative exchange between SCS and the Project Proponent to gather additional information for review and examination. This exchange includes Findings—New Information Requests (NIR), Non-Conformity Reports (NCR) and Opportunities for Improvement (OFI)—that are issued by SCS to the Project Proponent. The Project Proponent must respond to NIRs and NCRs in order for SCS to render a verification opinion. At this time, all findings

have been appropriately addressed by the project proponent and subsequently closed by SCS.

The Findings from the verification of the Darkwoods Project are compiled in Appendix 1- “List of Findings” to this report.

3 Verification Findings

3.1 Remaining issues, including any material discrepancy, from previous validation

This is the first project verification. At the time the verification audit was initiated, the validation of the methodology under the double approval process was still underway. Additionally, several corrective action requests from project validation remained open. During the audit, changes were made to the project and methodology to address these issues. These issues were all closed prior to the completion of the verification audit. No issues remain open that impact the status of the verification reported here.

3.2 Project Implementation

During the site visit and by review of project documents, SCS verified that the project had been implemented as described in the validated project design document, with the exception of the deviations to the monitoring plan described in section 2.2 of the monitoring report. These deviations include:

- 1) The network of field plot measurements installed does not meet the requirements of the methodology because not all analysis units were sampled. SCS approved this deviation because the audit team determined that the effect of not sampling some analysis units was not likely to lead to a material misstatement in the total number of VCU's issued to the project during the current monitoring period.

Under the selected methodology, the net carbon impacts of the project are determined by a comparison of model runs representing the baseline and project scenarios. A network of field inventory plots is used to assess the accuracy of the modelled project scenario and apply a confidence deduction if the model does not accurately predict actual project-scenario carbon stocks. The project achieves carbon benefits by avoiding harvest that would have occurred in the baseline scenario. In the case of the current monitoring period, the harvests that actually occurred in the project scenario and the harvests that were assumed to have occurred in the validated baseline scenario are limited to 9 analysis units. Five of those analysis units were sampled, and these five analysis units represent 98.19% of the total project area from which project carbon benefits are derived for the current monitoring period.

The audit team determined that monitoring of analysis units that were not changed in either the baseline or project scenario was not relevant to the number of VCUs claimed in the current monitoring period. Additionally, because the model error term that is derived from the on the ground plot network is calculated using an area-weighted approach, the audit team determined that the difference between actual standing stocks and modelled stocks for the 1.81% (by area) of the analysis units from which carbon benefits were generated but sampling did not occur would have to be very large to constitute a material error in the total number of VCUs generated. Based on the observed differences between model predictions and actual on-the-ground carbon stocks in analysis units that were sampled, the audit team determined that a model error in the un-sampled analysis units large enough to materially affect the number of VCUs verified was very unlikely. Because the audit team found the consequences of this deviation to be unlikely to lead to a material misstatement in the number of VCUs issued, and because the methodology specifically allows for the plot network to be developed over a period of up to 5 years, the deviation was approved.

Additionally, because project monitoring was carried out in winter conditions with several feet of snow on the ground in the project area, the project proponents were unable to measure lying dead wood. As described in section 2.2 in the monitoring report, the plot data were compared against the modelled carbon stock data as if lying dead wood had been included in the sample (i.e. project scenario lying dead wood was assigned a value of zero in the monitored plots while making model-field data comparisons). The audit team agrees that this represents a conservative comparison because: (1) if the model were over-predicting lying dead wood, the assumption that the carbon stock in lying dead wood on each of the monitoring plots would cause an increase in the model error component of the uncertainty factor, while an under prediction of the carbon stock in lying dead represents a conservative estimate of the project's net GHG benefits. Hence, the deviation was approved.

- 2) The carbon stock monitoring program has not yet been integrated with other standard operating procedures and project documentation maintained by the project proponent. However, clear monitoring procedures for the current monitoring period were presented that were found to be consistent with the methodology. The audit team did not find the lack of integration of monitoring procedures with long term project documentation to be a cause for concern at the first verification.

3.3 Completeness of Monitoring

As described in section 3.2, the project proponent has requested deviations from the monitoring protocol described in the PDD and methodology. For the reasons

outlined in section 3.2, SCS determined that the deviations did not materially affect the conservativeness of the application of the methodology. Consequently the deviations were approved. Parameters that were included in the PDD's specified monitoring program (as described in Table 8 of the PDD), but were not measured are $V_{LDW,c}$; $L_{i,t}$; $d_{n,i,t}$; $N_{i,t}$; and $D_{LDW,c,i,t}$. These parameters are all related to the estimation of carbon stock in lying dead wood. Lying dead wood was not measured due to snow at the time of field work, but was instead assumed to be zero. Because, under the methodology, field measurements of lying dead wood are used only to assess model error, and because the assumption that lying dead wood is zero increases the model error in the event that the model over predicts the carbon stock in that pool, omission of these parameters was deemed conservative.

SCS reviewed spatial inventory data related to project implementation during an office visit. Subsets of data were compared to aerial imagery available for the project area to confirm the accuracy of polygons delineated in the spatial database. Additionally, the verification team compared mapped stand boundaries to field observations in the project area during a helicopter overflight of the property and confirmed general correspondence between the spatial database and field conditions.

When SCS initially reviewed project documentation, no network of field plots had been installed since the forest inventory was established in 1999. SCS determined that the methodology required actual carbon stocks be measured in the project area at a maximum interval of 5 years, and issued a non-conformity report for the lack of recent field sampling. In response, the project proponent installed a network of inventory plots. This network did not meet all of the requirements of the methodology, but was approved by SCS for the reasons outlined in section 3.2.

Data obtained from measurements made on sample plots within the project area was assessed by re-measuring two of the initial twelve inventory plots established by the project proponent. SCS calculated the carbon stock in the two inventory plots using the methods given by the methodology and compared the carbon stock from SCS measurements to that calculated by the project proponent. The results in both plots were within 5% of the SCS biomass estimate. SCS observed that this was consistent with internal quality control checks carried out on different plots by the project proponent. The audit team considered this sufficient data for reasonable assurance that the data collected during project monitoring was of sufficient accuracy.

In total, the PDD and the selected VCS approved methodology describe a list of 24 data and parameters to be monitored (Table 8 of the PDD). The verification of these data and parameters can be summarized as follows:

Parameter	Reported in	Verification method
	GIS database	Comparison of polygons to aerial imagery; helicopter flyover

	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
Mean Tree Age	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Verified PDD. Not monitored at this time, as monitoring of mortality requires multiple measurements over time.	NA
	Darkwoods Carbon Model Ex-post 2008-2010 v8.7.xlsx	Review of scaling reports
	Zero value applied in both project and baseline case as described in PDD.	Conservative, as more harvest occurs in the baseline than project scenario.
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	Verification cruise, review of calculations
	Parameters relate to lying dead wood measurement and are not reported. See deviation discussed in section 3.2	NA
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	review of calculations
	Darkwoods Sample Plot Data Mar 2011 v3.2.xlsb	review of calculations

3.4 Accuracy of Emission Reduction Calculations

SCS reviewed all *ex-post* calculations used to derive the estimate of the project's net emissions reductions. This included a review of the following calculations:

- calculation of carbon stocks for the measured inventory plots
- comparison of measured carbon stocks for sampled analysis units to those predicted by the model to estimate the model error component of the uncertainty factor
- calculation of sampling error used to estimate the inventory error component of the uncertainty factor
- combination of model error and sampling error to estimate the overall uncertainty factor
- calculation of net project emissions and carbon storage in wood products from *ex-post* harvest volumes
- comparison of project emissions to verified baseline scenario emissions to estimate net emissions reductions
- calculation of market leakage risk discounts
- calculation of buffer account contributions
- calculation of VCUs

For most of the calculations, the project proponent used the same software and spreadsheets that were used to generate the *ex-ante* estimates of project net greenhouse gas reductions and removals reported in the project document. SCS did not discover any errors in the calculations used to derive the project's net emissions reductions. All calculations had been carried out in a manner consistent with the approved methodology and verified project design document.

3.5 Quality of Evidence to Determine Emission Reductions

Data and parameters used to determine net emissions reductions were derived from either measurements taken in the project area or were documented in the approved methodology or PDD. Three main types of *ex-post* data were collected that SCS verified:

- spatial inventory data, which was assessed via (a) a comparison of polygon boundaries to available imagery in the project area during a meeting with the GIS analysts responsible for maintaining project data and (b) a helicopter flyover of the project area, during which mapped stand boundaries were compared to field conditions;
- plot inventory data, which was assessed with a verification cruise; and
- actual harvest volumes, which were assessed by a review of scaling reports.

SCS did express some concern regarding the quantity and quality of field data used to estimate the project's uncertainty factor. Initially, no recent carbon inventory plots were in place from which to estimate the uncertainty factor. After issuance of NCR2011.1 the project proponent installed a network of 12 plots. SCS found the

sample size to be small when compared to the project's total forest area and noted that the inventory clearly did not meet the methodology's (1) requirement that every analysis unit be sampled, and (2) guidance of establishing enough plots such that the estimate of carbon stocks across all polygon or analysis units will lie within 10 percent of the true value of the mean at the 90-percent confidence level. As a result, SCS issued NIR2011.4. At the time of the issuance of NIR2011.4, the methodology was not yet approved. In response, the project proponent clarified that the methodology allows for the plot network to be developed over a period of five years before approaching targeted precision levels, and that the precision levels specified are guidelines, rather than requirements. The language of the final version of the methodology was altered to make this clearer and was approved by two different validation bodies in the VCS double approval process. Additionally, in an email from Carolyn Ching of the VCS to Todd Frank of SCS dated April 20, 2011, VCS indicated that *"the plot installation, at this time, is in line with the rules and requirements of the VCS and the methodology."* Consequently, SCS accepted the existing sample size. The initial plot network showed an inventory error of 32.4% and a model error of -20.5%, suggesting that, on average, the model underestimated carbon stocks in the project area.

Additionally, SCS found the initial sampling scheme employed to be biased because when the initially selected plot location did not match the expected forest type based on existing inventory, plots were either not measured or moved to a location that more closely matched the existing inventory. As a result, SCS issued NCR2011.2. In response, the project proponent provided a corrective action plan to mitigate the risk of sampling bias that included installing additional plots in locations that were initially omitted from sampling. SCS accepted the corrective action plan and verified the inventory on site and through data analysis after the additional plots had been installed. After installation of these additional plots, the model error component of the uncertainty factor rose to -25.2%. However, due to the increased sample size, the sampling error component of the uncertainty factor fell to 25.8%. The final project error was calculated as 0.6%, the sum of the inventory error and model error components. The methodology allows for a negative value for the model error term (i.e. an under prediction of carbon stocks) to compensate for error introduced by sampling. Because the methodology specifies that a minimum uncertainty factor of 1.5% be applied to all projects, the final uncertainty factor was 1.5%.

Overall, SCS found that, after issues raised by the audit team had been corrected, the quality and quantity of information used to determine the project's net emissions reductions met the requirements of the selected methodology.

3.6 Management and Operational System

SCS assessed the management and operational system of the project through interviews with the project proponent and technical service provider and review of project documentation. SCS observed that the project proponent, the Nature Conservancy of Canada, has a thorough system in place for documenting standard operating procedures for data collection and analysis and keeping project records.

However, as stated in section 2.2 of the monitoring report, procedures specific to project monitoring have not yet been incorporated in this documentation. SCS did find that existing procedures for the current monitoring period were adequate. The project proponent provided SCS with data from a check cruise undertaken on forest inventory plots taken by an independent team for quality control purposes. Review of this data confirmed that data collected in the initial inventory was accurate. These results were consistent with data collected by SCS through re-measurement of a separate set of inventory plots. Interviews with those responsible for project implementation, as listed in section 2.5, including technical responsibilities such as GIS analysis and carbon modelling, revealed a high level of experience and competency. SCS came to the conclusion that the management and operation system in place for the project is suitable.

3.7 Non-Permanence Risk Rating

SCS used the VCS “Tool for AFOLU Non-Permanence Risk Analysis and Buffer Determination” (referred to as VCS Risk Tool below) as the basis of the assessment. The following section outlines the SCS audit team’s findings in relation to the requirements set forth by VCS.

Generic Risk Factors (VCS Risk Tool Table 1)

Risk Factor	Self Assessment Risk Rating	Findings
Risk of unclear land tenure and potential for disputes:	Low	Review of title documents confirm clear title to land. Low risk assigned.
Risk of financial failure:	Low	Review of financial documents for the project show that revenue generated from the project is capable of supporting the project’s ongoing costs. The project proponent demonstrated to the audit team that a credible financial plan is currently in place. Low risk assigned.
Risk of technical failure:	Low	The primary project activity that results in carbon benefits is a reduction in harvest volumes. There is little opportunity for technical failure in such an activity. Further, the project management team has demonstrated extensive experience in forest management. Low risk assigned.
Risk of management failure:	Low	The team responsible for project implementation has extensive experience in all facets of project management, as confirmed via interviews during the on-site audit. Low risk assigned.
Risk of rising land opp. costs causing reversal of sequestration/protection:	Low	The land is owned by an organization with a long history and clearly stated mission of conservation. Past organizational history does not support a

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		significant risk of reversal based on rising opportunity costs. Low risk assigned.
Risk of political instability:	Low	Project is clearly in an area of low political instability. Low risk assigned.
Risk of social instability :	Low	Project is clearly in an area of low social instability. Low risk assigned.
Risk of devastating fire:	Low	Fire is clearly an important part of the ecosystem, and fires do occur in the project area on a regular basis. However, literature referenced by the project proponent clearly establishes the fire return interval as greater than 50 years. As part of project implementation, steps have been taken to mitigate the impact of fires that do occur, including a fire protection agreement with the BC government and a manned fire tower within the project area. Because of the project's size and diverse species composition, the audit team, in consultation with local technical experts, determined that a "devastating" fire would be a rare event. Finally, the models used to derive project carbon benefits already include a factor used to adjust for annual disturbance events. Low risk assigned.
Risk of pest and disease attacks:	Low	The project area, as well as much of British Columbia, is in the midst of a major pest epidemic. Insects are likely to have a significant impact on the project area. However, this has been accounted for in the modelled estimates of project scenario carbon stocks that are used to derive the project's net emission reductions. Consultation with local technical experts confirmed that the species diversity in the forest mitigates the risk of devastating disease impacts. Additionally, management of the project area includes activities designed to mitigate the impact of insect attacks, including salvage logging. While insect impacts are likely to continue in the project area, the audit team determined that they are unlikely to be catastrophic and that there was adequate evidence to support a low risk level.
Risk of extreme weather events (e.g. floods, drought, winds) :	Low	Consultation with local technical expert confirmed that a low risk of reversal due to extreme weather events is appropriate for the area.
Geological risk (e.g. volcanoes, earthquakes, landslides):	Low	Consultation with local technical expert confirmed that a low risk of reversal due to geological risk factors is appropriate for the area.
Overall Risk Rating	Self Assessment:	Verifier Assessment: Low

	Low	
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IFM Risk Factors (VCS Risk Tool Table 6)

Risk Factor	Self Assessment Risk Rating	Findings
Devastating Fire Potential	Low	Low risk assigned. See discussion above.
High Timber Value	Low	Although the project area does have valuable timber, the land is owned by an organization with a long history and clearly stated mission of conservation. Past organizational history does not support a significant risk of reversal because of high timber value. The project proponent has shown a credible financial plan for the property that is not dependent on significant ongoing harvest. Given that extensive harvest would be counter to NCC's stated mission and that the project's financial projects are sound, a low risk level is assigned.
Illegal Logging Potential	Low	Timber markets in British Columbia were demonstrated to be well regulated, with little regional presence of illegal logging. Low risk assigned.
Unemployment Potential	Low	The region has a diverse economy with many local employment alternatives to commercial timber harvest. Project has elected to ramp down harvest levels over the first several years of the project to reduce the impact on local communities. Low risk assigned.
Overall Risk Rating	Low	Low

VCS requires that projects be given the overall risk rating associated with the highest risk factor from the analysis above. The highest risk rating from the categories above is "low", so a low risk rating is assigned to the project.

The specified buffer contribution range for low risk avoided logged to protected forest projects is 10-15%. The VCS program update from 13 April 2010 specifies that the required buffer withholding percentage shall be the maximum percentage in the buffer range for the determined risk class, unless justification for a lower withholding percentage can be demonstrated. The project proponent presented an application

of the VCS Version 3 Risk Assessment Tool, which was in draft form at the initiation of the verification audit, but which was approved by the VCS before the audit was completed, as justification for a lower withholding percentage. This tool led to a total risk rating of 7.5%, which is less than the minimum withholding percentage given by the VCS 2007 tool. Consequently, the audit team determined that a 10% buffer withholding was appropriate for the project.

4 Verification conclusion

SCS was able to arrive at an opinion regarding the accuracy of the calculated emission reductions and removals from the Darkwoods Forest Carbon Project through a review of the Project Design Document and the supplementary documentation, additional requested information, a site visit, and an office visit with the project proponent. SCS confirmed the soundness of the data regarding project eligibility, inventory procedures, baseline and project characterization, methodologies related to the calculation of carbon stocks and GHG reductions and removals. Through the risk based verification assessment, SCS has determined that the Darkwoods Forest Carbon Project, developed by the Nature Conservancy of Canada (project proponent), Ecosystem Restoration Associates, and 3GreenTree (project developers), is in conformance with the Voluntary Carbon Standard and meets the minimum quality standard.

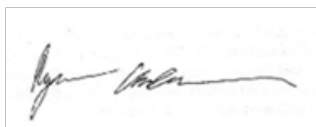
Furthermore, all issues identified during the verification were resolved and found to be in conformance with VCS standards. The monitoring report and data are considered accurate, complete, transparent, and free of material misstatements. Therefore, SCS is able to issue a positive verification opinion for the 956,865 metric tonnes of CO₂e emission reductions and removals by the project for the monitoring period from April 1, 2008 – December 31, 2010. While all 956,865 VCUs will be issued, 10% of the project's CO₂e emission reductions and removals (107,849 metric tonnes CO₂e) will be set aside in the VCS's buffer pool. As such, the project will be issued a total of 849,016 VCUs for Emission Years 2008-2010. The VCUs in this Verification Report are consistent with those claimed by the project proponent in the 2008-2010 monitoring report.

Reporting period: *From April 1, 2008 to December 31, 2010*

Verified emission reductions in the above reporting period:

Year	Net Emissions Reductions (t CO2e)	Leakage Discount (t CO2e)	Uncertainty Discount (t CO2e)	Buffer Pool Contribution (t CO2e)	Net VCU to Project (t CO2e)
2008	317,676	38,038	4,195	31,436	244,007
2009	331,713	38,266	4,402	32,838	256,207
2010	439,332	40,979	5,975	43,576	348,802
TOTAL				107,849	849,016

Net emission reductions, after subtraction of buffer pool contribution: 849,016 t CO₂ equivalents

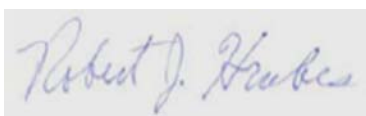


Name: Ryan Anderson

Title: SCS Contractor

Company: Contractor to Scientific Certification Systems

Date: April 28, 2011



Name: Robert J. Hrubes

Title: Senior Vice President

Company: Scientific Certification Systems

Date: April 28, 2011



Forest Project Verification

Appendix I: List of Findings

Verification under the Verified Carbon Standard

Reporter/Member:

3GreenTree / ERA / Nature Conservancy of Canada

Project:

Darkwoods Forest Carbon Project

Non-Conformity Reports:

NCR Number 2011.1 dated 3/11/2011

Finding:

The selected methodology requires that a network of field plots be established to assess the accuracy of modeled carbon stocks. No field plots have been installed in the project area.

Proponent Response:

A total of 16 field plots have been installed after this NCR was issued; with 2 plots being check cruised. One of these plots was a replacement (see sampling bias NCR), leaving data from 15 plots. These plots provide field sampling representation of 98.3% of the Analysis Units affected by baseline and project activities during the monitoring period. The data and related analysis have been provided, and are included in the list of reference files shown in the Monitoring Report, Section 5 – Supporting Files and Additional Information. The verification team was able to conduct field audits of the plot sampling at the site visit.

Auditor Response: A network of plots was installed by the project proponent in response to this NCR. Initially, that plot network was found to not fully meet the requirements of the methodology, which had not yet completed the double approval process at the time this finding was issued. See NCR 2011.2 and 2011.4. After some conservative deviations proposed by the project proponent, which are discussed in section 3.2 of the verification report, a response to NCR 2011.2, and clarifications to the methodology, which were then double approved by other validation bodies, SCS determined that the proponent's response adequately addresses the finding in accordance with the VCS standard and approved methodology.

NCR Number 2011.2 dated 3/11/2011

Finding:

The sampling design used in the installation of field plots was found to be biased because when the initially selected plot location did not match the expected forest type based on existing inventory, plots were either not measured or moved to a location that more closely matched the existing inventory.

Proponent Response:

3GreenTree proposes the following Corrective Action Plan:

Although we have full confidence in the unbiased field decisions of the registered professionals undertaking the measurements, we recognize statistically there are 2 possible sources of plot selection bias:

1. due to not installing plots on certain available site options due to stand type;
2. due to moving the targeted UTM coordinate to achieve a plot representative of the sampled polygon.

Our plan to mitigate the possible affects of selection bias is to install additional randomly selected plots in the locations where a possible bias might be impacting biomass results (continued next page).

So our proposed plan to remove or reduce the possible plot selection or location bias is:

1. Reviewing the 13 proposed plot locations which were not selected for a plot installation, the following rationales were provided:

i. 2 locations were not used because the pine stand was mis-typed based on age (younger than expected) or disturbance (old selective logging)

ii. 2 locations were mislabeled for target leading species (1 fir/pine vs. pine; 1 HCF/PL vs. target P)

iii. 9 locations were balsom/spruce leading versus the expected spruce/balsam. These stands should be very close in biomass content, however the crews followed the instructions explicitly in looking for a spruce leading stand. So, we will randomly select and install a sample of these unselected plot locations. We will stratify this selection based on the 3 categories above to increase the probability of fully representing the possible bias conditions. Our plan would be to randomly select and install at least 1 new plot in each category (minimum 3 total).

2. After reviewing with field crew, it turns out that of the 12 plots installed, the following plot moves were made:

i. 3 plots were moved <40m to find a suitable live center post (an unbiased minor relocation)

ii. 2 plots were located in the same polygon, so were moved to meet proximity constraints (we allowed plots in the same polygon as long as they were >150m apart). The proximity constraint makes these essentially randomly located.

iii. 6 plots were not provided a UTM centroid, only a sample polygon. The crews located the plot at the first accessible point that met the proximity constraints. We view these as essentially randomly located within the polygon (in particular, the plots were not moved due to stand type or other features).

iv. 1 plot was moved 249m from the target UTM to locate at a spot the crew felt was representative of the average polygon condition.

So, of the 12 plots, it appears 1 plot (category iv) has a potential of selection bias, where the crews moved the plot to find a more representative site within the stand. We propose to locate a plot on the original UTM (at the closest suitable live centerpost) for inclusion in the dataset.

So in total, our plan would be to install at minimum 4 additional plots, located randomly, but targeted at the strata above. The intent would be to provide remove or significantly reduce the potential plot selection and location bias in the original plot network.

We are looking for guidance on the suitability of implementing this plan to close this NCR.

As per the above plan, 3GT/NCC undertook to install an additional randomly selection of 5 plots within the list of potential plot locations which were reviewed, but not installed in the original plot layout.

Five (5) additional plots were installed, in addition to the 12 plots were installed originally. One new plot served as a replacement for the original plot, resulting in a total of 16 plots being used in the ex-post monitoring analysis. Two (2) plots were blind check cruised.

1. The original Plot #108(103)-280 was moved in the field to a location the field crews judged to be representative of the stand, which unintentionally created a possible source of bias. This plot was replaced with a new plot at the original polygon centroid UTM. The original plot and data were discarded and not used in ex-post analysis and calculations.

2. Four (4) plots were randomly selected (from office generated random numbers) within the 13 plots locations which were identified but not sampled by the field crews during the original plot installation. The available selection list of 13 plots was stratified to capture representation of the reasons the plots were not installed originally and to assure representation within the most important analysis units. In the reference file "Darkwoods sampling plot data Mar 2011 v 3.2.xls", the 13 plots identified as possibly biased are highlighted in red, and the additional installed plots highlighted in green. The random selection data is also provide in the lower section of the spreadsheet.

As per the Methodology/PDD, 2 of the 16 plots (plot #22 and #25) fell within a single polygon, and such these plots were combined into a mean value and counted as a single sample observation. Therefore the final sample size (N), was 15.
The results of these data from 15 plots were used to calculate the uncertainty factor, as per the methodology and PD.

Auditor Response: The additional sampling has substantially reduced the risk that the estimated model error was biased because of plot selection choices. SCS spent an additional day in the field observing the installation of the additional plots and found the accuracy of measurement to meet the requirements of the methodology and standard. Proponent's response adequately addresses the finding in accordance with the VCS standard and approved methodology.

New Information Requests:

NIR Number 2011.3 dated 3/11/2011

Finding:

Please provide copies of the following documents:

- Project Monitoring Report
- Approved Methodology
- Completed Validation Report

Proponent Response:

The Monitoring Report and additional data reference files have been provided to SCS.

The PD validation report and approved methodology have been provided to SCS.

Auditor Response: The project proponent supplied the required documents. The proponent's response adequately addresses the finding in accordance with the VCS standard and approved methodology.

NIR Number 2011.4 dated 3/11/2011

Finding:

Section 13.3.2 of version 8.6 of the selected methodology requires that the sampling design target a precision of +/-10 percent of the true mean at a 90% confidence level. The sample size used in the current field plot network was insufficient to achieve the targeted precision. Further, the methodology has not yet been approved, and language regarding the field sampling requirements is understood by the audit team to be in flux. Please provide written clarification of the plot installation sampling requirements, including precision requirements, as approved by the methodology validators.

Proponent Response:

The 3GT/ERA IFM-LtPF methodology (VM0012) does not specifically mandate a number of plots or sampling intensity, nor specify a plot accuracy result requirement. Instead the methodology lays out a process which results in an uncertainty factor calculation that penalizes the project credits based on the level of sampling error (a combination of model error and inventory error) in each verification period. A key

assumption in the methodology is that inventory and model error are strongly affected by sampling intensity, and a small number of plots would result in a high level of error, and hence a higher uncertainty factor penalty. This process was used within the methodology to provide flexibility to the project to install the complete carbon plot network over several years for efficiency, cost and feasibility reasons (and to be able to use carbon revenue to fund plot network development). We felt it would be rare for any larger scale project to start with a full carbon plot network, and that most would want installation staggered to have an even annual workload on re-measurement. At the same time, the uncertainty factor calculation provided a method to ensure the calculations are conservative, regardless of plot numbers.

Specifically, the areas of the methodology which are relevant to the plot sampling expectations can be found at:

1. **Section 8, Page 15** - Starting inventory requirements: these requirements ensure projects have a minimum level of starting inventory accuracy, such that ex-ante estimates are reasonable (i.e. a project without any inventory data could not proceed without first establishing starting inventory requirements).

2. **Section 13.2.2, page 77** - Number of Plots, Precision, and Sample Size: *“The proponent will develop a plot network with representation in every polygon or analysis unit³⁷ (based on the primary stratification method identified in Section 8 and used throughout the project calculations) and a design target of establishing enough plots such that the estimate of carbon stocks across all polygon or analysis units will lie within 10 percent of the true value of the mean at the 90-percent confidence level³⁸.”*

³⁸ *The uncertainty factor calculation in Section 11.4 accounts for, and penalizes the project credits for higher uncertainty error, and hence this target is provided as guidance for plot network design to achieve the lowest uncertainty factor. “*

This was intended to be guidance for how the plot network is to be designed overall (i.e. a target). This is not phrased as, nor intended to be, a sampling intensity threshold or mandatory requirement. IF interpreted as a mandatory threshold, then the uncertainty factor calculation would be redundant and would not have been included (the uncertainty factor would never vary).

“For practical purposes it is recognized that on large or complex project areas the plot network may need to be developed across several years (no longer than 5 years) to approach the target level of precision. Samples sizes should be evaluated for suitability following the initial monitoring period and then adjusted as appropriate to achieve the desired level of precision.”

This clearly indicates the expectation that the plot network will take time to install. Again, it does not specify further.

3. **Section 9.2, page 32** – 9.2 Determining Actual Onsite Carbon Stocks. This section provides guidance on the steps to take to incorporate monitoring data (including plot data into the ex-post calculations). This does not specify a number of plots or sampling intensity.

4. **Section 11.4, page 55** – 11.4 Calculation of an Uncertainty Factor. This section repeats requirements on how plots sample data are utilized to calculate and uncertainty factor. Other than representation of each Analysis Unit, no specific plot number or sampling intensity is provided.

Darkwoods Deviation:

The monitoring deviation requested by Darkwoods is we cannot provide 100% representation of all analysis units as required by the methodology. Instead we have sampled 98% of the analysis units where project or baseline activity occurs in the monitoring period (i.e. we sampled analysis units where harvesting occurred in the baseline and the project). We were unable to access the remaining analysis units prior to spring, however these analysis units do not affect ex-post credit calculations (the baseline and project scenario would be affected by

inventory errors equally and offset).

We have established 15 plots across 6 analysis units (of 15 total AU's), which provides the capability to calculate a solid uncertainty factor. We expect our error is higher than it will be as additional plots are installed, so we actually took the risk of heavily penalizing ourselves in this first monitoring period (turns out our modeling was very conservative and underestimating volume, which offset some of the inventory error caused mainly by low plot numbers). Additional comparable evidence was provided from a BC MOF inventory accuracy assessment of the surrounding approximately 1.0 million hectares of crown land with similar inventory methods, which demonstrated a significantly lower inventory error than our current plot data. We also tested actual project harvest volumes from provincial log delivery reports against predicted (modeled) merchantable volume by area for 3 years, and demonstrated a very close prediction vs. actual (which demonstrates the model error was lower than our current data). This corroborating data is available in the PD, Appendix 4. So, we are being conservative, with the uncertainty factor accounting for a higher deduction than would be expected from additional comparable evidence.

To summarize, the methodology does not require a minimum number of plots, and our uncertainty factor calculations from the 15 plots can be shown to be conservative against comparable data.

Update (April 18, 2011) – The final wording of the final methodology and PD was consistent with original wording outlined above, with the intent that the plot network target the $\pm 10\% @ 90\% CI$ over time, with an uncertainty factor deduction calculated from actual plot results statistics. VCS and Rainforest Alliance (the 2nd Methodology Assessor, and PD Validator) have confirmed this interpretation is consistent with their assessment of the Methodology and PD, and viewed as an acceptable plot sampling level given the uncertainty factor calculation method.

Auditor Response: It has been clarified that the original methodology language had been intended as guidance, rather than a requirement. The final language in the methodology, which has now been double approved, is more clear with regards to sampling intensity guidance. Additionally, SCS received guidance from VCS that indicated that the existing plot installation is in line with the rules and requirements of the VCS and the methodology (email from Carolyn Ching to Todd Frank dated April 20, 2011). Consequently, SCS determined that the proponent's response adequately addresses the finding in accordance with the VCS standard and approved methodology.